

Active Inference BookStream #001.09 ~ "Governing Continuous Transformations"

BookStream | 43:51

Hello, it's February 7th, 2023. We're here in ACTIV bookstream number 1.09. Off to you, Blue.

Hi, I'm Blue Knight. I am a researcher in New Mexico, and we are the Active Inference Institute, a participatory online institute that is communicating, learning, and practicing applied active inference. You can reach us at our social media links here. And just a reminder, this is a recorded and an archived live stream, so please provide us with feedback so that we can improve on our work. All backgrounds and perspectives are welcome here, and we will be following good video etiquette for live streams. And just as we're mentioning that all backgrounds and perspectives are welcome here, we will be having our final discussion coming up in about four weeks. So if you would really like to participate, if you've watched some of the live streams and have some questions, we would really like to get some outside perspective. So please get in touch with us so that you can participate in our final discussion.

So like I said, if you want to participate, this is how. And so we are still discussing governing continuous transformation. And today our goal is to discuss section nine, so chapter nine, and we will be doing book map roadmap and all of the typical things. So I already introduced myself. Daniel just gave us the date. Why don't you introduce yourself?

It's Daniel in California. I'll pass to Tyler.

I'm Tyler in California. I am a DAO researcher and protocol designer.

All right. All right.

So here we are at the end of part two. So next week, the author is going to join us for a discussion over part two, the free energy governance section of the book. And then we'll be moving into part three. I think we're going to do two chapters at a time as we go through part three. And I don't think we'll be discussing the engaged scholarship interviews, but if you're curious about them, I encourage you to read the text.

Okay.

Tyler, do you want to read this first part of the abstract for us?

Sure. All right.

So beneath all chaos is an underlying structure. Our challenge is to exploit disorders underlying order. To survive, living organisms are encoded to resist disorder, i.e. entropy, by way of self-organization. Firms are nothing more than molecules and are subject to obeying the very same laws of thermodynamics. Entropy will eventually rise.

Three specific propositions previously developed as part of this book's underlying dissertation research are outlined.

One structure, the stronger affirms cross hierarchical top down bottom up cognitive throughput coupling between

and within governance and operational channels. The less dominant affirms top down approach, cultivating a

form of CEO centurism

and strategy ownership. Blue, you want to take the next next one?

Yes. So two, cognition. The stronger affirms cognitive disposition toward environmental enactment and innovative abduction as opposed to adaptation.

The stronger the firm's strategic control and the smaller the risk of falling for the financial control trap.

Three, capabilities. The stronger the board's DMN-centric duality stewardship, empowering upper echelon to sustainably balance exploitation versus

exploration as well as financial versus strategic control. The stronger the sustainability of successful and continuous strategic renewal and hence the firm's prospects for survival.

Strategy is less macroscopic but more subatomic. Less linear and deterministic but more quantum and discontinuous.

There is a dominant tendency of glorifying the realist beauty of top down linear cause effect relationships to the very detriment of quantum self organization.

And I'm going to start off just reading this part from the book. So Bijan says, I'm trying to see it over my screen.

Order anywhere in the cosmos is so singular, it always requires an exemplary mechanism or a process. While randomization requires no, I can't read the word.

I can't read the word. It is simply the way of the world. When particles are allowed to act randomly, whether it's the contents of a glass on of club soda on the rocks or the countless atoms making up the air in the room, they slam into each other, exchanging energy until complete randomizations of their positions and velocities is observed.

And so Bijan quotes this Lanza et al. 2020 paper. And I was just like thinking of like firms as like organizations of people.

And like it really brought this picture to mind of like, you know, atoms randomly slamming into each other, like people randomly like slamming into each other, like the mosh pit versus like the board meeting.

And so like, I just wanted to bring this like juxtaposition of like, like, what is it to like minimize free energy in terms of like assembling humans as opposed to like assembling atoms. Anyway, I just had to share this like mental image.

Okay.

Okay.

Tyler, do you want to read this one for us?

Sure.

So beneath all chaos is an underlying structure. Our challenge is to exploit disorders underlying order. To survive living organisms are encoded to resist disorder, i.e. entropy, by way of self organization.

Firms are nothing more than molecules and are subject to being the very same laws thermodynamics.

And so yeah, I thought this was, I thought the connection here that was interesting was comparing, saying that the laws of thermodynamics apply to firms. And that's what's really going on.

And we talked about this in previous streams. I know we have, Blue's going to talk about a little more in detail here about whether that's more of like an analogy comparing to thermodynamics or something like a real underlying structure of thermodynamics.

And we'll talk about that more shortly.

That also brings to mind like the mosh pit. And like, I wonder in our base case as humans, like, are we like, just chaotic? Like, is our natural inclination to be like increasingly chaotic and like we override that?

It just, I don't know. I had all these mental images as I was reading this as well. So, yeah, Tyler put this in. So I love this. I'm going to read it because I love it.

So, uh, so, uh, so Bajan says, Oh, he actually quotes Gould at out of 2016. So, um, and in the quote, he says, if physical theories were people, thermodynamics would be the village, which over the course of three centuries, she smiled quietly as other theories rose and withered surviving major revolutions in physics, like the advent of general relativity, relativity and quantum mechanics.

The other theories find her somewhat odd, somehow different in nature from the rest. Yet everyone comes to her for advice and no one dares to contradict her.

Einstein, for instance, called her the only physical theory of universal content, which I am convinced that within the flame framework of applicability of its basic concepts will never be overthrown.

Her power and resilience lay mostly on her frank intentions. Thermodynamics has never claimed to be a means to understand the mysteries of the natural world, but rather a path toward efficient exploitation of said world.

Um, yeah. Do you want to add comments here, Tyler?

Tyler Lowe Yeah.

I mean, something I liked about this was that like the kind of like mystical business leader, like a Steve Jobs character is kind of like a trope in the business world.

And it's kind of like almost like a joke at this point, but there's also like a real case for the mystical business leader, which is almost kind of what I think Bajan is like describing here is that, you know, the real state of the world of like operating, um, as a firm in a very uncertain, uncertain is probably not the word I should use, but like, you know, the real state of the world of like operating, um, as a firm in a very uncertain, uncertain is probably not the word I should use.

But like in active, in an active sense, like is a very mystical, receptive way of engaging with the world. Um, and I really liked making that connection of mysticism with like business, like, you know, corporations.

Tyler Lowe One other angle there is organizations are doing work. They're doing hopefully useful work.

Tyler Lowe

And it's like, well, if all the air molecules just slammed against the wall, it would fall over. Or if they all formed into a droplet, you'd get rain in the room. Like thermodynamics helps us understand what is likely to happen given those interacting state spaces. And again, interestingly, it has to do with extracting work, useful work from this background of just vibration that's not work.

Yeah. And the other thing I'd like to just to build on that is like, it's okay to not know what's going on. Is that like, that doesn't necessarily need to be a source of stress. It's just like, you can harness those energies and kind of see which way the wind is blowing, so to speak, without necessarily understanding all the underlying principles. And like, that's totally okay. And I think that's generally something that gives a lot of people, both in the business world and even in our personal lives, a lot of stress.

But this is also, I think, a very like freeing way to approach an uncertain world.

Yeah. And there's also like, just to come maybe to close the loop or maybe open it up a little bit more. There's also the prospect of doing work that you don't know is going to be fruitful. Right. And so like that is this like

uncertainty minimization, like you could spend, you know, your whole, like an hour trying to crack open a nut and the nut is empty. Like that's also possible. Right. So like doing, even though you're doing this, you're doing this, you're doing this, you're doing this, you're doing this, you're doing this, you're doing this, you're doing this.

Like, that's also possible. Right. So like doing, even though you're doing work like this, maybe useful work, it might not bear fruit. So there's, there's that like uncertainty element, like as just a human or also like as a firm.

So I just wanted to bring back these slides, like while we're talking about thermodynamics, we talked about this a long time ago and we talked about the difference between like thermodynamic entropy and information theoretic entropy.

And so I just wanted to kind of bring this back and just remind everyone that there's a difference. There is relationships between the two that are still being hashed out like mathematically and not totally understood.

But, but the second law of thermodynamics is that this entropy, like disorder in the universe is always increasing.

And like here, like the, you know, the energy is always lost in a transfer in, in work as heat to the environment in, yeah. So less and less usable energy becomes available. All right.

All right. And then in information theory, we kind of talked about entropy as like with an inverse relationship with information.

So here you can see like a, a, a event that is low probability is high information.

So like if you're decoding a stream of letters, I always like to think about it like this, cause it was explained to me like that.

And it made so much sense. If you're decoding a stream of letters and, and like, you're expecting a message and this was Claude Shannon.

Um, and kind of like how he, or how it was explained to me, how, how he kind of came up with this.

So you're decoding a message and you get the letters T H it would be really freaking weird if like the next letter was Z.

Like, I don't know any words that have THZ. Like, like that's a very low probability event. Right.

So, so it would be like a very high information thing or like, you know, maybe something came before the TH and you missed it, but there's some probability like T H E would be great.

Or like T H I, but like THZ wouldn't make any sense unless like the TH was at the end of some previous word.

And so like, you can kind of deduce what to expect in those kinds of situations.

Um, and so, yeah, the, um, probability distribution is like when there's 50, 50 probability.

So this is the second graph on the right.

When there's 50, 50 probability, like the flip of a fair coin, that's super high entropy.

Like you don't have any idea if it's going to be heads or if it's going to be tails.

Um, but like here it's low entropy.

If it's like a, a coin with two heads, right?

There's heads on both sides.

Like you can only come out with heads.

So like, there's very low entropy in such a coin flip because the outcome is always heads.

Um, right.

If there's no distinction between the two sides.

So I don't know, that's the, the thermodynamic entropy versus information theoretic entropy.

And I think we talk a lot about thermodynamic entropy, but maybe forget that, um, the FEP is really kind of founded in the information theoretic entropy as well.

Um, do you guys have anything to add here?

No.

Okay.

Um, all right.

So Bajan makes this claim in this section, and I'm not going to read the claim, but I'm just going to point out that here we kind of have a really concise definition of like, he's been emphasizing the structure, cognition, and capabilities for a few chapters now.

And so here he defines structure as the, the circular copper coupling, circular coupling of governance predictions and operational stimuli channels in the form of prediction error minimization.

He defines cognition as environmental engagement in the form of enactment rather than adaptation.

And he defines capabilities as sensing and sense making in the form of duality management.

And so this chapter was really kind of a good, um, like summary chapter.

I thought I can, and kind of gives us a chance to go back and revisit like a couple of the concepts that, that, um, we talked about before, but maybe need to highlight one more time.

Um, do you want to read this one Tyler for us?

Sure.

So yeah, through the rest of the chapter, Bajan is going through each structure cognition, um, and he's really kind of going deep and kind of recapping what he's been covering through the previous, the rest of the book.

Uh, so for structure, he says the stronger firms, cross hierarchical top-down bottom-up cognitive throughput coupling between and within governance and operational channels, the stronger the firm's orientation towards free energy, surprise minimization based on self-organization.

Um, the less dominant firms, top-down approach, cultivating a form of CEO centrism and strategy ownership, the stronger the sustainability of successful and continuous strategic renewal and hence the firm's prospects for survival.

Well, thanks.

Um, and so this is kind of like the, the punchline of the structure.

And so I pulled out maybe just a couple of topics from, um, the, from before the punchline, uh, to talk about.

Um, and, and so this kind of like top-down bottom-up coupling, I was like looking for a picture of like, what does top-down bottom-up coupling like look like?

Like, how do you see that?

Like, how do you see that?

Can I visualize that?

Actually, I was like looking for a picture to put on the previous slide.

And I came across this like, what 2008 paper from emotion resonance to empathic understanding, a social developmental neuroscience account.

And like my background is neuroscience, but it's definitely like cellular and molecular neuroscience.

So I don't really have a good, like comprehensive reading of, of social and developmental like neuroscience and what that looks like.

And I was like, I loved this figure because it just was so like encompassing of so many of the things that, um, we've talked about so far.

Daniel, have you read this paper?

Have you ever seen this before?

Like I hadn't either.

And I came across the figure and I was like, oh, I have to get the paper now.

Um, and so I, it was nice to kind of just delve into this and it's really about, um, what's required for empathy.

Um, but I just, so I'll, and I'll talk more about it, but here you see like the top-down bottom-up coupling.

Um, so it's bottom-up information processing, top-down information processing relation to like with really relative to the individual and then the context or situation.

Um, and, and there's this perception action coupling and then like 2008, I'm like, the perception action loop from 2008.

This is cool.

Um, and, and they talk about shared representation, um, and this like self other awareness, which we talked about a couple of book streams ago.

Like, um, how like distinguishing the, the self versus the other self and non-self.

Um, so I thought that this was really cool and executive functions, regulation and control, metacognition, like the looking at, um, like that's like double loop learning.

The, the metacognition aspect.

So I thought that this paper was really cool.

And, and, um, it really was like, wow, all of this is, is like grounded in empathy.

Um, anyway, it was, it was neat.

It was a good read.

Uh, it's destiny and Meyer 2008.

And the, the link is up there if you want the DOI or any of that stuff.

Um, and it made me think like all of this is an empathy, grounded in empathy.

So like, what is like, we know like, or at least it's thought in business that like to be successful as a business person, your emotional quotient, your EQ is like way more important than your IQ.

Like your ability to relate to others and interact with people.

So like a good emotional, high emotional quotient versus like an intelligence quotient is way more crucial to, um, business success.

And so I just wonder about like the relationship between, if any, between free energy governance and like the emotional quotient.

Like, is there like, I like, because I never talked about this directly in the book, but it just made me think like, how important is empathy for this kind of governance structure?

Um, I don't know if you guys have any thoughts here on this.

Yeah.

I mean, I think empathy is just really like sensitivity to your environment or sensitivity to like other actors and their internal states.

And so in the context of a firm, like it doesn't necessarily need to be like their, the, you could think of an emotion necessarily as like for a, another firm, uh, maybe the internal state of this external other firm.

So maybe like you are dependent on this other firm, things aren't going the way you would expect them to go with this other firm maybe, or there's a backlog or their API is being kind of intermittently down.

Maybe that is like a signal into the internal state of that firm and the internal state of the ecosystem as well.

And so I think that that might be like analogous to an emotion in a human context, maybe.

Yeah.

They had a bunch of, um, definitions actually in the paper.

Um, and I'm, I can't pull it up cause I'm, I'm screencasting.

So I, I'm not going to let you guys like go through my, go through all my 25,000 open windows, but maybe we'll pull them out.

Um, and, and this is like maybe a good question for Bajan about the relationship between empathy and the free energy governance when we get to see him next week.

But they had like several different definitions of empathy too, in the paper, which I thought was interesting.

They really, really tried to take like a very like logical approach to it.

And it was cool.

Um, okay.

So here is a claim from the paper.

Bajan says the higher the organizational friction in terms of obstructing bottom up error messages, traveling to and reaching top down prediction generators, the higher free energy, the potential for surprise.

And the more likely that top down dominant logics fossilized prediction models and hence reinforce affirms entropy.

Um, and like, this is something that maybe Tyler and I have brought up either on stream, but we've talked about it off stream a lot.

Like how do we reduce this organizational friction?

Like where are the, um, examples to that show us like, what is the best way to like reduce this friction to like create a good top down bottom up coupling?

Like we know that this coupling is super important, like at all layers across the hierarchy, but like, can we find like an empirical way like that works in all situations to build that, that structure?

Um, and, and that, that I think is like a challenge or maybe a future research question.

I guess another thing that is a little counterintuitive about this too, is that oftentimes the things that will reduce friction and maybe like the longer term feel high friction in that moment.

So for example, like in previous streams, we've been talking about like dissonance as being like a way of thinking about that friction, right?

Well, the easiest way to deal with dissonance, the lowest friction way you could deal with it personally, just like not thinking about it and just like tunnel vision and just like ignore it.

And don't worry about these weird feelings that you have.

Um, and that, but that obviously doesn't help the friction of the organization as a whole.

Um, and so, yeah, there's this kind of, especially in your personal life too, by analogy is like confronting your own emotions that you don't want to think about in the short term, really easy to just like power through and just keep doing what you're doing.

But if you can like process these and integrate these emotions, um, that's a lot healthier for you long-term.

So it's an interesting kind of like tension between like your individual tendencies of like not wanting friction and like friction in the organization as a whole.

Also to connect back to the thermodynamics, what is friction?

Well, it's when a force is applied on some surface or through some media and it results in heat, which is dissipated, taking away energy from what could have been done for work.

And now think about friction engineering.

Do we coat every object with Teflon?

Do we want everything to just at the lightest touch fly off into space?

So just because we can increase or reduce friction actually doesn't tell us, especially over timescales, like you're bringing up Tyler, whether we want something to have a lot of friction, like where the rubber hits the road or whether there are some fundamental trade-offs with how much heat the tire is losing in contact with a smooth road versus how much grip it has on a loose road.

And so all these different trade-offs are where the actual art and science is going to come into play.

But friction is not just a term that's used every day to describe annoying communication patterns.

FEG is pointing towards that potentially being understood as an actual physical force.

Do you want to read this one for us, Tyler?

Sure.

So the second proposition cognition.

So the stronger a firm's cognitive disposition towards environmental enactment and innovative abduction as opposed to adaptation, the stronger the firm's orientation towards free energy surprise minimization based self organization.

The stronger the firm's strategic control and the smaller the risk of falling for the financial control trap.

The stronger the sustainability of successful and continuous strategic renewal and hence the firm's prospects for survival.

And so I'm going to read another quote from this section on cognition.

And I brought back this picture because I think it's important.

Okay.

So Bijan says,

Cognition in the form of world enactment is the purpose directed action centric and social construction of shared meaning and shared expectations anchored in the understanding that a firm's environment is a matter of environmental engagement in the form of interaction revealing significance and assigning probabilities to a bounded set of external unknown states.

And so I'm coming back to this paper and I'm looking at the information processing bottom up top down.

And here I see this shared representations and we've seen this in this figure in this paper, they're talking

about body language and somatic response, heart rate increase or lower and all these things that have been measured in the generation of empathy.

But like, is empathy maybe something that is required for shared meaning or shared expectation?

Like, I wonder, like, do you have to have that?

Like, like, is a sociopath, like devoid of empathy, like living in a world where like things don't mean the same thing and expectations aren't the same?

Like, maybe, right?

Like there might be a strong link there that I wonder if it's been explored or not.

So I'm just curious if you guys have thoughts on what developing shared meaning and expectation is and if it requires some kind of empathy with others.

Yeah, I share that feeling. One of the big questions I've had through this and honestly, it's on a personal level. I've always been interested on like whether you could build tooling that would help create that shared meaning and expectations or whether this is really just constrained by like the individual.

Like if you're if you have people who are sociopaths in your organization, maybe it's just like there's certain the way that they're wired kind of limits the way that they can engage and create shared meaning with the rest of the organization.

And so, yeah, I'm kind of curious what you all think about that, too.

Like, I don't have an answer for it, but maybe that'd be a good question for Sean.

All right. Daniel, thoughts? No thoughts?

No thoughts? No thoughts?

Well, to have interpersonal empathy as a behavior, as a feeling, that's one whole area.

But in the creation of organizations, it's almost like we have to ask not how strongly we feel or what we feel, but whether some future person in that organization outside of our direct perception, cognition, action loop is going to have some sort of outcome.

And that's the outcome, because I think it's easy just to sketch out that you could feel really empath and empathetic or any emotional state, and it could result in a totally discordant state, even for somebody you care about or vice versa.

So how do we not get lost with just the introspection of the feeling of empathy and translate that into systems whose efficacy is equivalent to what we would have intended?

But that may not feel like empathy all the time.

Yeah, I mean, I feel like, especially over the last couple chapters, we keep running to like a similar concept of like, or theme of how the individuals wired in terms of like, you know, empathy and their ability to create shared meaning, and how that bubbles up into the rest of the organization.

And yeah, like, I keep feeling like, there's no way out of like, hey, we're just limited as humans in our ability to like, integrate external information, and that limits the ability of the organization, but sounds like Daniel was maybe sounding more optimistic now like, well, actually, maybe you can build some, maybe there's tooling and systems you can build that can have organizational empathy, even if the individual doesn't have it.

So that brings to light, like something that Daniel and I have talked about before, working on like, the discussion of memes, like randomly random insertion random plug for the eco meme project.

But like, when you're really thinking about painting a picture to convey like a shared meaning, like we talked

about the echolocation, how the dolphin, like literally can like, paint a picture like I saw a fish over here, and can like translate like, puts the picture of the fish into the other dolphins mind, like there's the picture of the fish.

Right.

And so like, you literally can paint the entire picture.

So like, even if we could do something as remarkable as that, we convey only the semantic content, like if you can say like, the fire hydrant is red, right?

And like, you guys both know, like can picture in your mind, like a red fire hydrant, right?

But like, we don't know what semiotic content is contained in that message.

Like, maybe someone's colorblind, maybe they don't know if it's red or green, like they can't tell.

And so like the fire hydrant appears in this gray thing that they think might be red.

Or maybe like someone has been in a house that burned down and the fire like the hydrant didn't work and they have this like trauma inspiration moment, right?

Like, so the semiotic content, the symbolism behind the message or the picture that you painted, even in a situation where you can perfectly create like the tooling to come to the shared meaning or representation.

Like you're never going to get the symbolism that exists for someone else.

It's a highly personal thing.

Like we can have shared symbolism too, culturally for sure.

But there's a deep level of personal symbolism in every, in every human that's dependent on your previous life experience, I think.

Ah, so coming back to abduction, because we did talk about it in this section.

I can't remember if it was in the last quote or not, but I just wanted to point out, this is a recycled slide, but coming back to abduction where Bajan talks about it is, you know, abductive reasoning.

This is a quote from the book.

Abductive reasoning usually begins with a surprising observation or experience.

This is what shatters our habit and motivates us to create a hypothesis that might resolve the anomaly.

Abduction is an inferential procedure in which we create a conjecture that if it would, if it would make the surprising anomaly part of our normal world of our normal understanding of the world.

So like this picture is like the best way for me to always think about abduction.

Like you can have a specific observation and you can come to a general conclusion, right?

And then you can have deductive reasoning where you have a general rule, but that's like leads to a specific conclusion.

That's always true.

And then when you have like an incomplete observation, like you can't see what's going on, you just go to the best guess.

And so in business and Bajan talks about this again and again and again, like you have to use abductive reasoning because we're never going to have a complete picture of like the past and future of the market, like the whole market, right?

Or even your target audience. So you're dealing with the best logical guess.

Oh, okay. Do you want to read this one for us, Tyler? This is you.

Sure. So the firm's generative model is the firm's eco niche.

If uncertainty and unpredictability dominate our worldly perception, well, then this reflects the very limitations and corrosiveness of our models and underlying processes generating those models.

There's no objective uncertainty. There's only your uncertainty. Environment is enacted.

Each act opens a new fork of multiple paths leading to two potentially very different outcomes.

Rally is subjective algorithm by an infinite number of worlds and superposition.

Our actions not only create the junctures leading to different paths, but we are in control to decide which path to take at each structure.

So, yeah, I thought this really summarized well the kind of inactive cognition.

So there's a proposition to someone proposition to here of FG and also just that was like a fun quote because we talked about a lot on the last stream.

But John's been using kind of quantum language here and we talked about that on last stream of K.

What exactly does this mean there by that?

I love your Excel spreadsheet quantum superposition of an Excel spreadsheet.

Like I love it.

It's great. It's capturing the late night blurry what action selection to take based upon the spreadsheet.

Great.

All right.

So, Bajan makes this claim also in this section.

He says, in fact, we are challenged not to allow for the human brain's conceptual limits to solely determine perception and sense making.

Instead, outsourcing some of the heavy information lifting to the environment in the form of extended cognition is the essence of optimizing prediction error minimization.

The more exponential and hence discontinuous market environment, the more we need self-organizing mechanisms that leverage the world as a supervisory signal challenging in circular causality the very perceptual content of our predictions.

And so, like we've talked a lot about cognitive offloading or extended cognition in terms of like cumulative cultural evolution, not in this bookstream per se, but like over the course of active inference, like from the very first beginning live streams, we've talked about it a lot.

Because, you know, we now are so much thinking is done externally, like in it's in cultured at some point, like with the smartphone, like we don't have to know how to get anywhere anymore.

We'll just map it. So I just am thinking like this self-organizing mechanisms that leverage the world as a supervisory signal challenging in circular causality, the very perceptual content of our predictions.

Like, I mean, I'm so amazed by ChatGPT.

I'm using it so much just because it's like, what does everybody know?

Like ChatGPT tells you what everybody knows. It is like the selective synthesis of all information. Right.

And so like if you know something that ChatGPT3 doesn't know, like that's great. You can write a blog about it.

But now like ChatGPT3, you can just write all the blogs about every other thing.

Because there's just like no, like it knows everything. Right.

It's got all of the information synthesis, except like the rare pockets of like knowledge.

So this is this this self-organizing mechanism, this ChatGPT3 like a global search synthesis information management system.

I don't know. What do you guys think about that?

Well, this is very out of my depth, both in terms of like machine learning and active inference.

I was thinking about this last night because this is like a huge debate right now.

It's like whether ChatGPT3 is actually showing like emergent intelligence or whether it's just like a clever, clever algorithm.

And I was thinking about this to you, like what are the differences between like ChatGPT3 and what I've been learning so far about active inference and like the crucial things that seems to be missing is like the inactive aspect of it.

Like it's not really in dialogue with this world. It's like stringing together information that's like a static. Right.

It doesn't even incorporate new information. Like if you were to ask it about the news yesterday, it couldn't it couldn't do that.

And it's not making it's not taking actions in the world and getting feedback. It's just like the static model of how the world works.

And so I don't know. It feels like fundamentally different to me. I'm not sure what that means.

I'm not sure if like something has to display active inference to have true intelligence.

I'm sure this is like a big, hotly contested topic. I'm sure you guys have many live streams about this.

But yeah, Daniel, Blue, I'm curious what you guys think.

What in the space of communication, the response it provides to you is an action that's not moving a mechanical arm on a table.

But in our Internet of everything, it's not so hard to see how the digital outputs from a generative algorithm do translate to physical objects moving.

But it may not even need to do that. It can send out messages on social media and definitely meriting further discussion as things go.

As things will continue to develop and gain more access to recent information, world model learning and action capacity soon.

So like I have a little bit of a technical response here.

I think that the question is not about true intelligence, maybe.

But the question that I hear talked about a lot is sentience.

Right. And so but like I read this big article about like is GPT-3 sentient?

And I think it was like from one of the inside Google people, the one that was released.

And now they're talking about it because everybody's using it.

And like sentient, they just said, like, you know, able to think and act independently.

But that's like not the definition of sentient.

Like, I'm not sure what the definition of sentient is, but it's not like autonomous.

Like they're using sentient and autonomous the same way.

And sentient is like able to feel things.

Right. Like, so I think like, let's look it up right now.

Let's define sentience for the world at large.

What does that mean?

What does that mean?

Because intelligent, sentient and autonomous are all different.

So hold on.

I'm looking on my phone because I'm screen sharing.

Words, words, words.

Humpty Dumpty, Alice in Wonderland.

People will use them how they want.

But that's everything.

So a sentient is able to perceive or feel things.

Right.

And so it's GPT-3 feeling anything.

I highly doubt it.

Perceiving a lot.

Sure.

But as to your thing about updating, Tyler, like I'm sure the model will update at some regular interval.

So like, just think about, you know, I mean, we're not there yet in terms of compute power, but maybe in 10 years, like it will be training on the entire Wikipedia catalog every single day.

So like, does Wikipedia have like a date entry for every day?

I think it does.

Right.

So like when, when you ask, like, what was the news yesterday?

It'll know that.

Like, that's just a matter of what's encoded and what's being processed through silicone.

It's only time.

Like we're, we're, we're way close.

So like, it will be able to update at some, with some regularity.

Like, will it update itself?

I don't know that that gets into like autonomous acting.

I don't know.

All those things are different.

I mean, that reminds me though, because we've had this discussion over the past 15 minutes about like empathy and do organizations have empathy.

Right.

And it's kind of like the same conversation as like GPT three of like, okay, well, like maybe organizational empathy.

Like it looks and just feels differently than like individual empathy.

Right.

Um, and so you can imagine the same, the same thing happening in GPT three.

And I don't know what that version of empathy would look like in GPT three, but maybe it's just like something that we can't quite wrap our hands around.

Um, it'll be awesome.

Again, I think a further discussion would be very timely, especially if, for example, communication from one person's syntax could move into a semantic space or a rhetorical space from the eco meme direction, as blue mentioned, and then be translated back into another semantic space for the listener.

That might be changing the amount of math detail or in a different human natural language.

So there's a lot of incredible opportunities.

Yeah, definitely.

I think, um, you know, it'll be interesting to see, to interrogate Bajan with such like esoteric,

I think to see, to interrogate Bajan with such like esoteric questions and see what he thinks.

Like, does a firm have empathy?

Like what, where did you even get this?

Well, we got it.

Sorry.

We're off task.

Um, anyway.

All right.

So moving on, uh, Tyler, do you want to read this one?

Preposition.

Sure.

So the stronger the board's DMN centric duality stewardship, empowering upper echelon sustainably balanced exploitation versus exploration, as well as strategic or financial for strategic control.

Stronger the firm's orientation towards free energy surprise minimization based self organization.

Stronger the sustainability sustainability of successful and continuous strategic renewal, and hence the firm's prospects for survival.

Cool.

Cool.

And so we talked about this last week a lot.

I didn't put a lot of extra information in here because we did like the antagonistic neural.

Um, that perspective last week.

Um, but I did just want to include the summary.

So Bajan at the end of the chapter summarizes it by saying in summary, the board must assume the role of sponsor or enforcer of three categorical imperatives defining the success and sustainability of strategic

renewal and firm survival.

First, foster cognitive coupling between and within all hierarchical layers.

I'm really excited to find out exactly how to do this.

Uh, second, embrace environment as action generated and socially constructed shared meaning and shared expectations, which we've talked about a lot today.

And third, de chain the firm from thinking in terms of antagonisms, but embrace real time duality management as a critical determinant of performance and survival.

Um, that's pretty much it.

Do you guys have any final thoughts?

The overtime period, the generative AI speculation moment, the hashtag 2023 questions that we all have.

Um, good times.

Um, so I, I, we, we talked about quantum, so I'm, I'm leaving us with this one, um, quote from this chapter, because it was just kind of like thrown in there.

He, Bajan says, classical linear cause effect thinking is no match for quantum exponentiality.

I wonder what he means by quantum exponentiality.

Like, are we again talking about position?

Are we talking about like, cause that's like, I guess the many states existing at once.

Is that like what he means by exponentiality?

What do you guys think?

I think Bajan has a poetic bent in him.

I'm not sure if we're meant to take this fully literally.

Exponents are used in a lot of quantum mechanics.

So there could be various readings just like any other good poem.

So we should jump in with these kinds of questions.

Um, cool.

Well, that's next week.

Um, so yeah, excited to wrap up this part and just a couple more book streams left for us, uh, winding down again, get in touch.

If you would like to participate in our final discussion, we would really welcome any external perspectives.

All right.

Thanks.

Um, let's see.